4(a)	compression/extension is proportional to force (provided limit of proportionality is not exceeded)	**B1**
4(b)	$(E) = ½Fx$ **or** $½kx^2$ **or** area under graph	**C1**
	$= ½ \times 8 \times 16 \times 10^{-2} = 0.64$ (J) **or** $= ½ \times 50 \times (16 \times 10^{-2})^2 = 0.64$ (J)	**A1**
4(c)(i)	$(E) = ½mv^2$	**C1**
	$0.64 = ½ \times 0.076 \times v^2$ $v = 4.1\ \text{m s}^{-1}$	**A1**
4(c)(ii)	$(\Delta)(E) = mg(\Delta)h$	**C1**
	$= 0.076 \times 9.81 \times 0.24$ (= 0.18 (J))	**C1**
	kinetic energy = 0.64 – 0.18 = 0.46 J	**A1**
4(c)(iii)	$v = 4.1\ \text{m s}^{-1}$	**A1**
4(d)	$W = Fs$	**C1**
	$d = 0.30 + (2\pi \times 0.12) + 0.25$ (= 1.3 m)	**C1**
	$F = 0.23 / 1.3$ = 0.18 N	**A1**